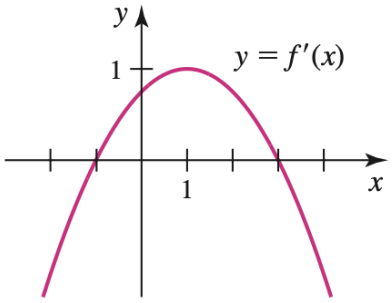
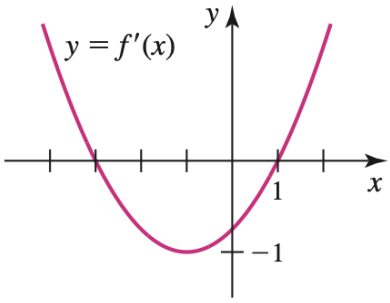
15–16. Functions from derivatives *The following figures give the graph of the derivative of a continuous function f that passes through the origin. Sketch a possible graph of f on the same set of axes.*

15.

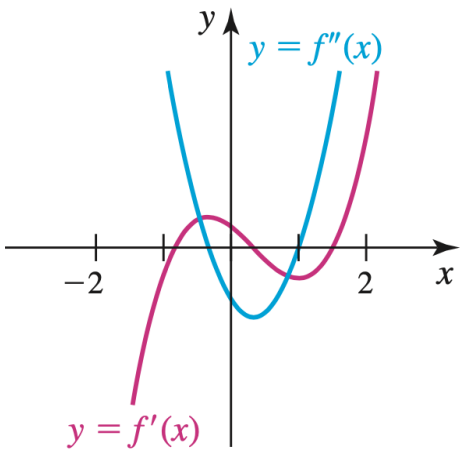


16.

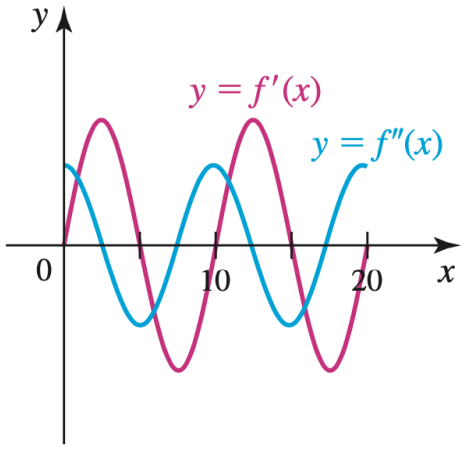


74–75. Functions from derivatives *Consider the following graphs of f' and f''. On the same set of axes, sketch the graph of a possible function f. The graphs of f are not unique.*

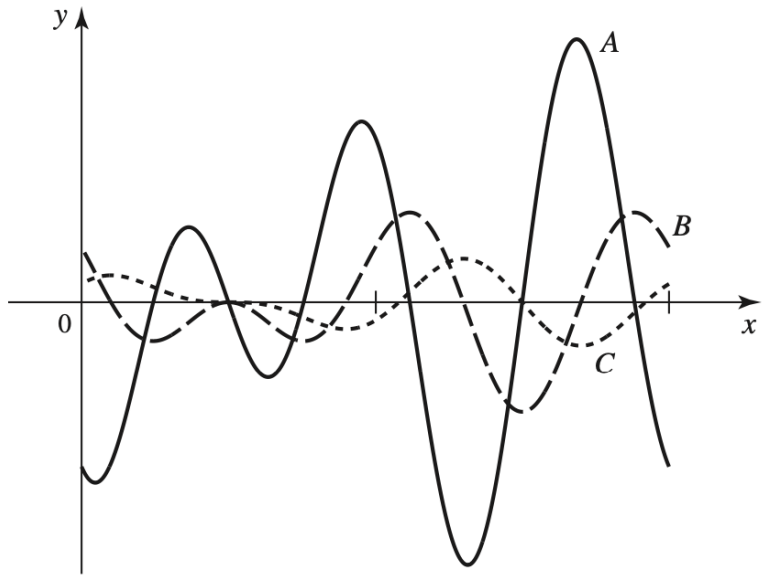
74.



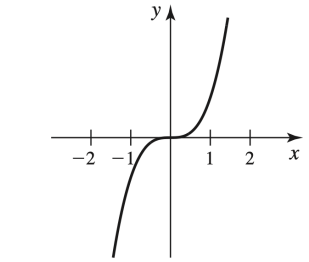
75.



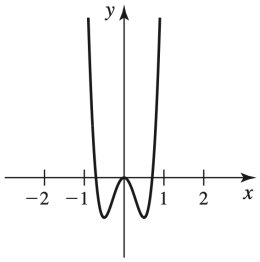
78. Graphical analysis The figure shows the graphs of f, f', and f''. Which curve is which?



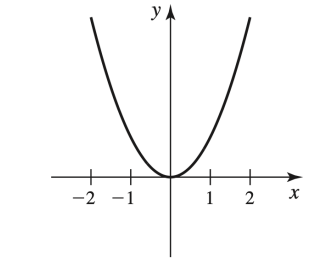
77. Matching derivatives and functions The following figures show the graphs of three functions (graphs a–c). Match each function with its first derivative (graphs d–f) *and* its second derivative (graphs g–i).



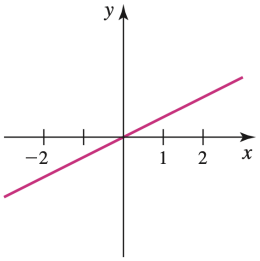
(a)



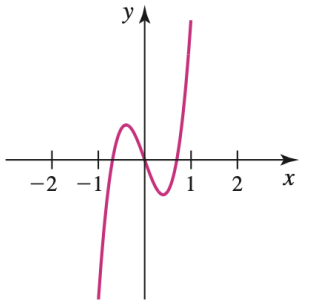
(b)



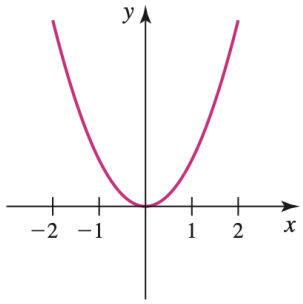
(c)



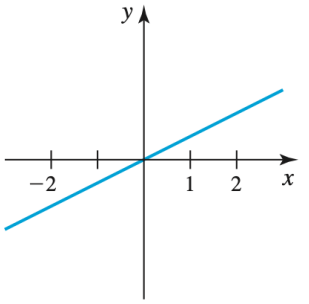
(d)



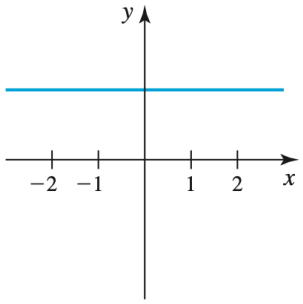
(e)



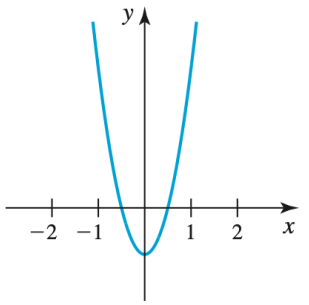
(f)



(g)

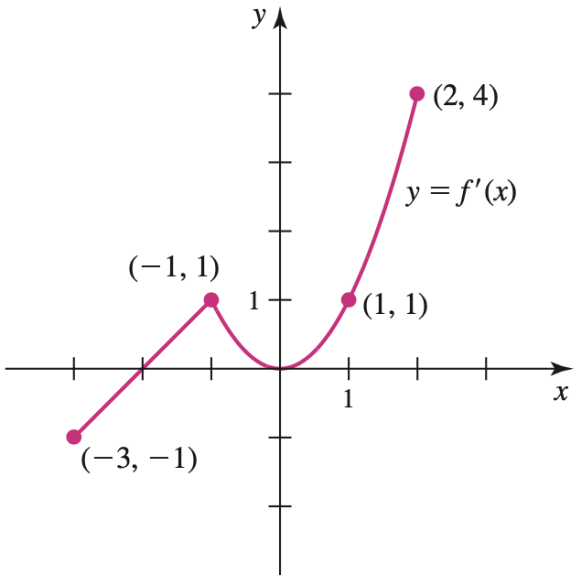


(h)



(i)

85. Interpreting the derivative The graph of f' on the interval $[-3, 2]$ is shown in the figure.



a. On what interval(s) is f increasing? Decreasing?
b. Find the critical points of f. Which critical points correspond to local maxima? Local minima? Neither?
c. At what point(s) does f have an inflection point?
d. On what interval(s) is f concave up? Concave down?
e. Sketch the graph of f''.
f. Sketch one possible graph of f.